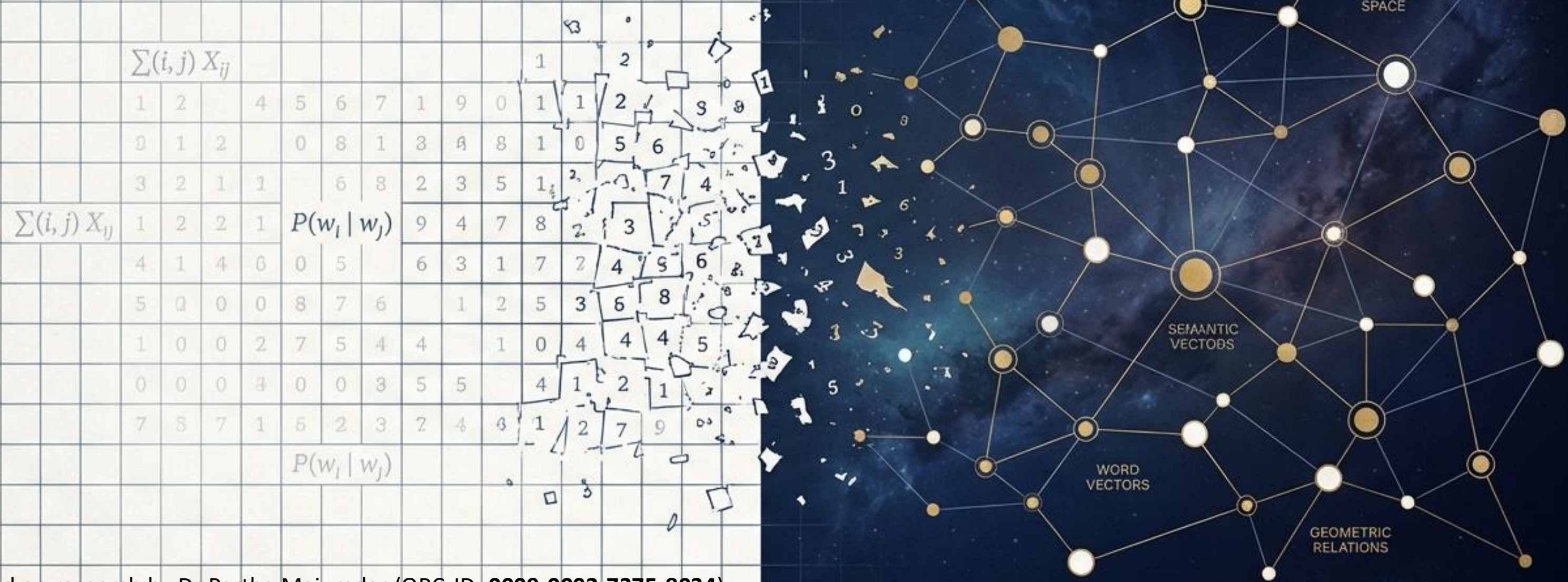


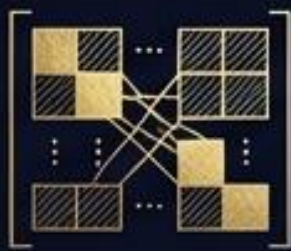
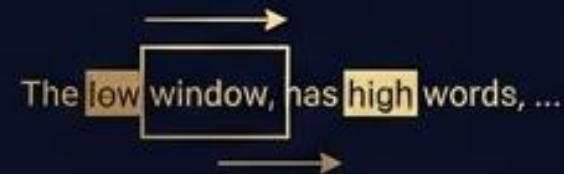
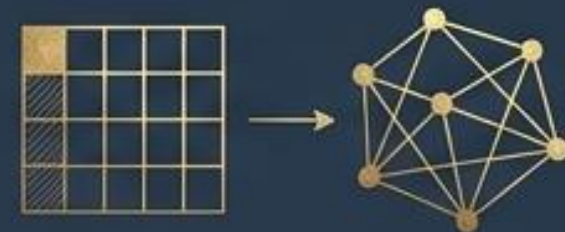
The Alchemy of Language

From Global Counts to Geometric Meaning: The Architecture of GloVe



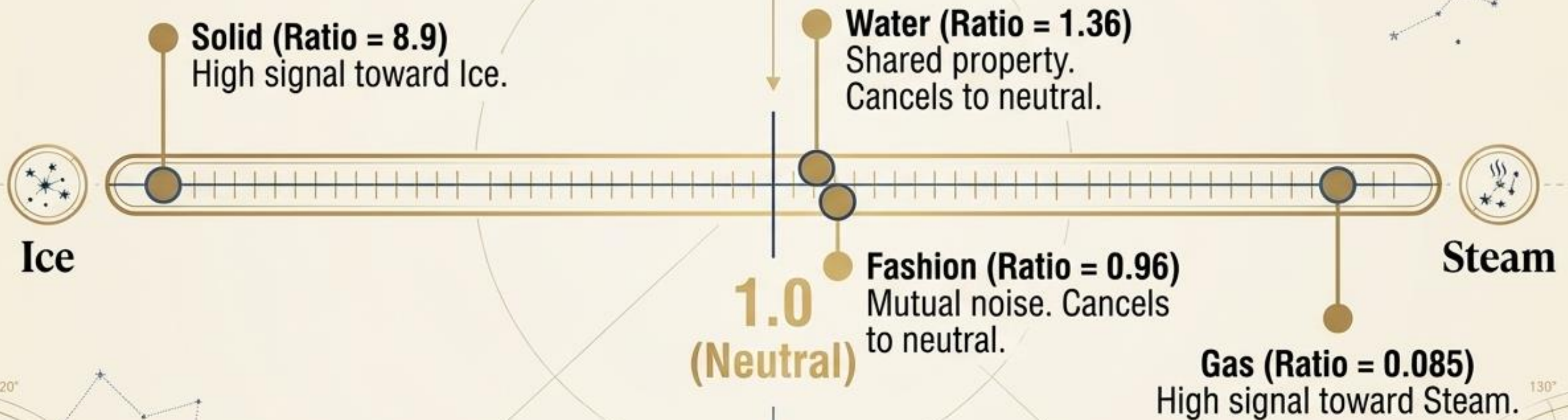
Based on research by Dr Partha Majumdar (ORC-ID: 0009-0002-7375-8034)

The Evolution of Embeddings

LSA (Latent Semantic Analysis)	Word2Vec (CBOW / Skip-Gram)	GloVe (Global Vectors)
 Context Scope: Global (Entire Corpus) Learning Approach: Matrix Factorization Core Strength: Robust statistical profile, but struggles with word analogies.	 Context Scope: Local (Sliding Window only) Learning Approach: Iterative Neural Prediction Core Strength: Captures linear arithmetic structures, but ignores macroscopic corpus statistics.	 Context Scope: Hybrid (Global Matrix + Local Windows) Learning Approach: Log-Bilinear Regression Core Strength: Merges global statistical rigor with the rich geometric structure of local neural models.

Noise vs. Signal Probability Scale

The Concept: Raw probabilities struggle to separate shared traits from mutual noise. Ratios perfectly cancel the noise.



Insight: Ratios extract discriminative meaning. Therefore, GloVe mandates that vector **differences** in the embedding space must equal the logarithm of overall co-occurrence probabilities.

The System Architecture

Stage 1: Co-occurrence Matrix

	1	2	3	4	
	0	4	5	6	
			3	2	4
	4	5	6	3	1
			6	9	1

Slide a window across the corpus to count proximity.

Output: Sparse global matrix X .

Stage 2: Embedding Lookup



Initialize random word vectors (W) and context vectors ($\tilde{W}$), plus scalar biases for each.

Stage 3: Interaction / Regression

$$W \cdot \tilde{W} \approx \log(X_{ij})$$

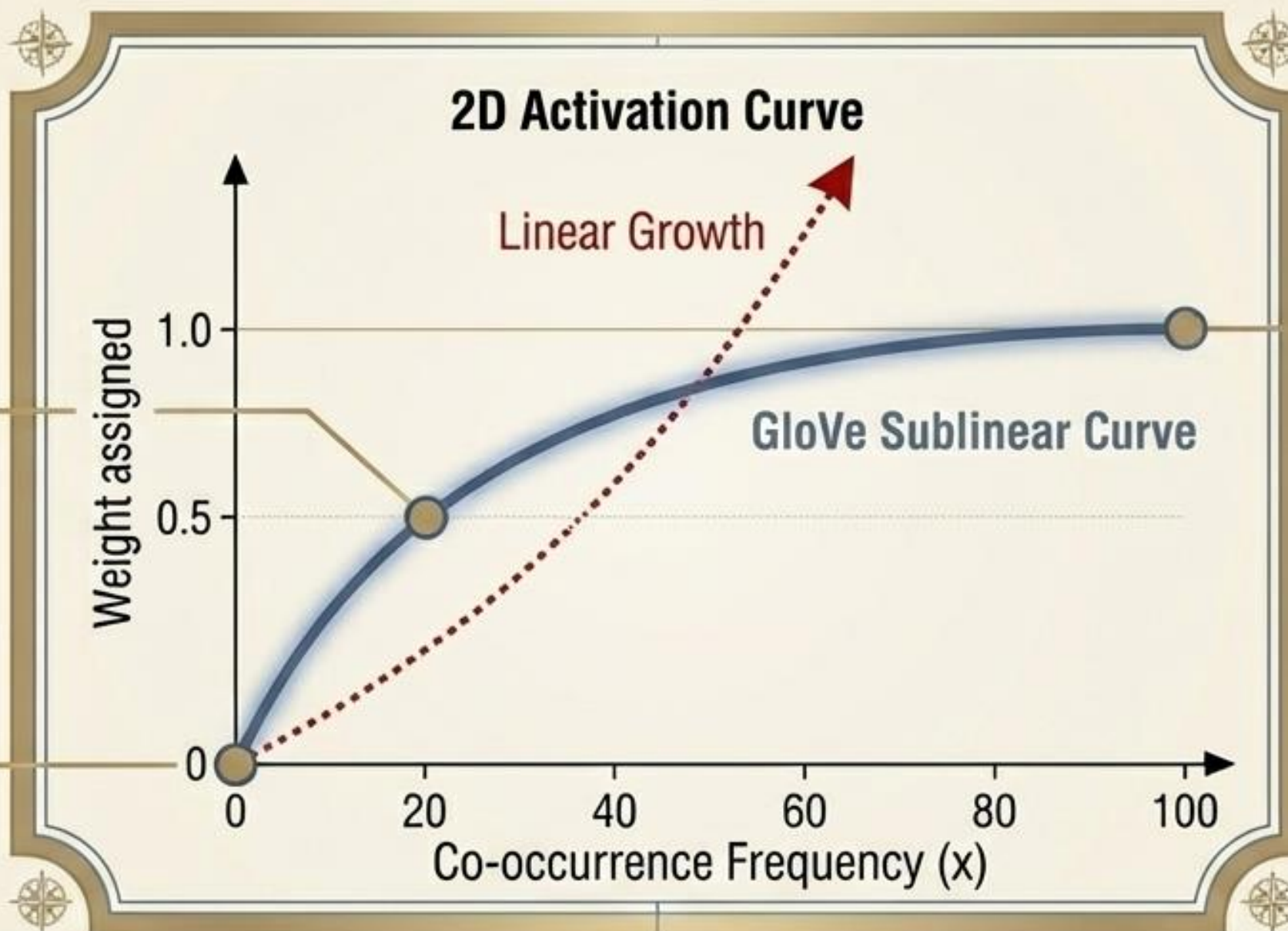
Compute the dot product of the word and context vectors.
Objective: Approximate the log of the co-occurrence count.

Stage 4: Error & Loss Update



Calculate the residual difference, apply the weighting function, and backpropagate to adjust vectors.

Control Panel



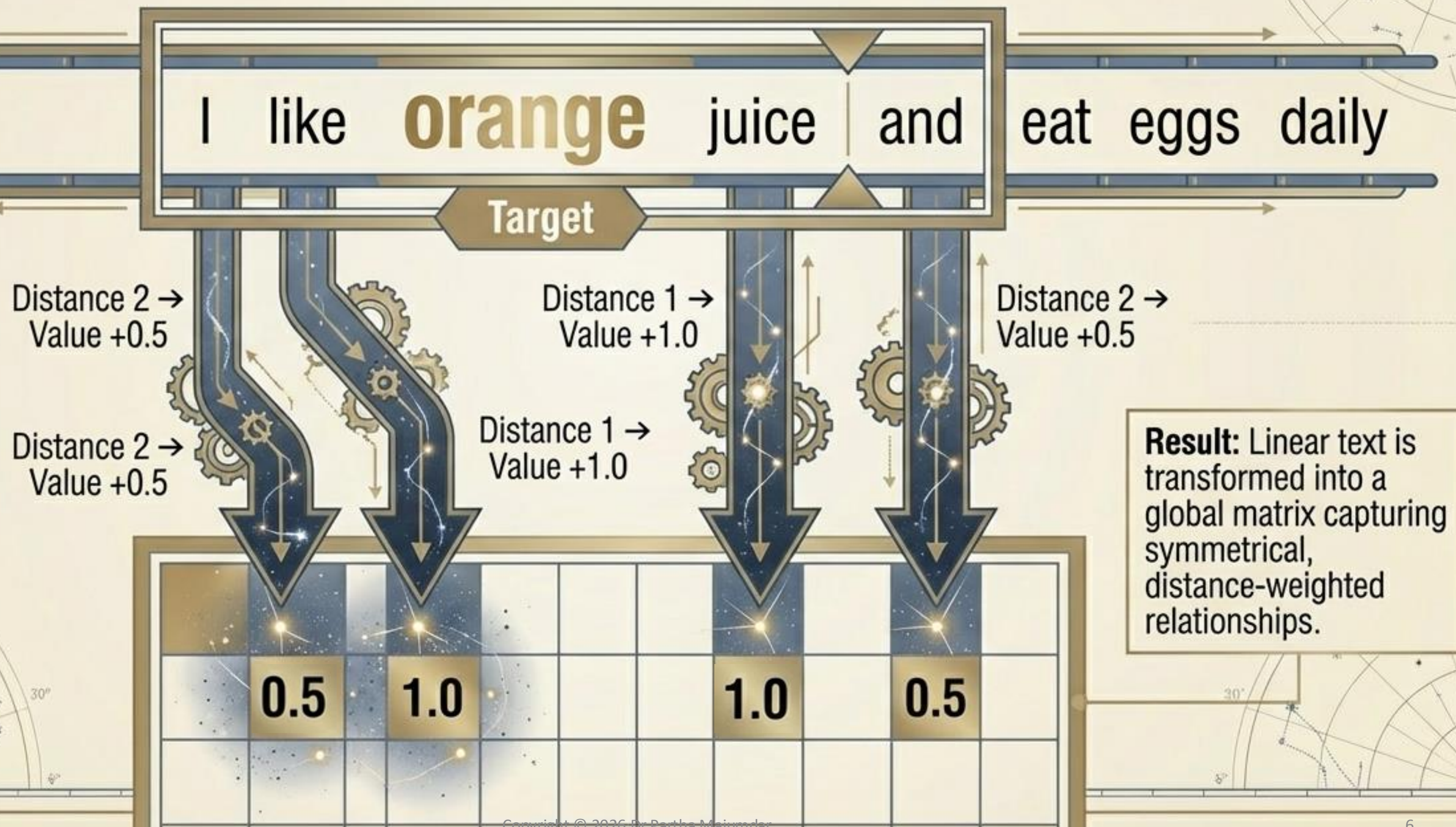
Sublinear Growth ($\alpha = 0.75$):
Shapes sensitivity.
Treats rare pairs cautiously; importance grows at a diminishing rate.

$f(0) = 0$: Ignores the massive sparsity (up to 95% zeros) of the global matrix.

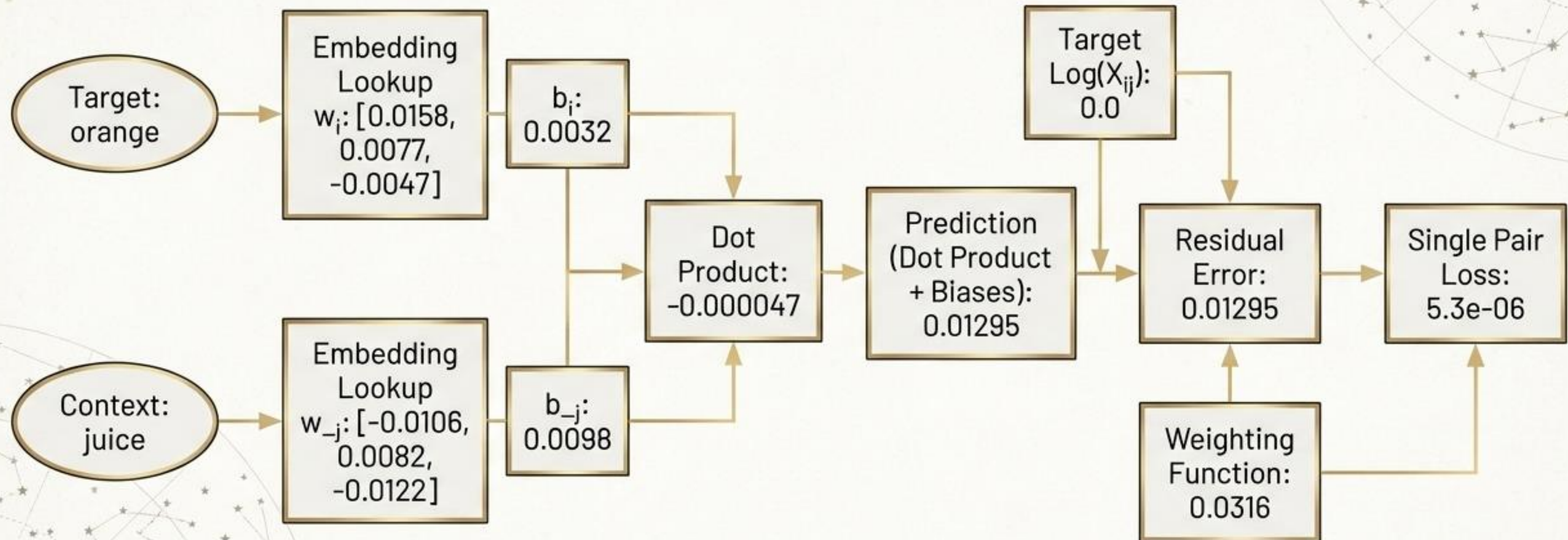
The Ceiling ($x_{\max} = 100$):
Plateaus to 1.0.
Prevents hyper-frequent stop words ('the', 'and') from breaking the loss function.

This weighting function isolates reliable signals—trusting frequent co-occurrences without letting them dominate.

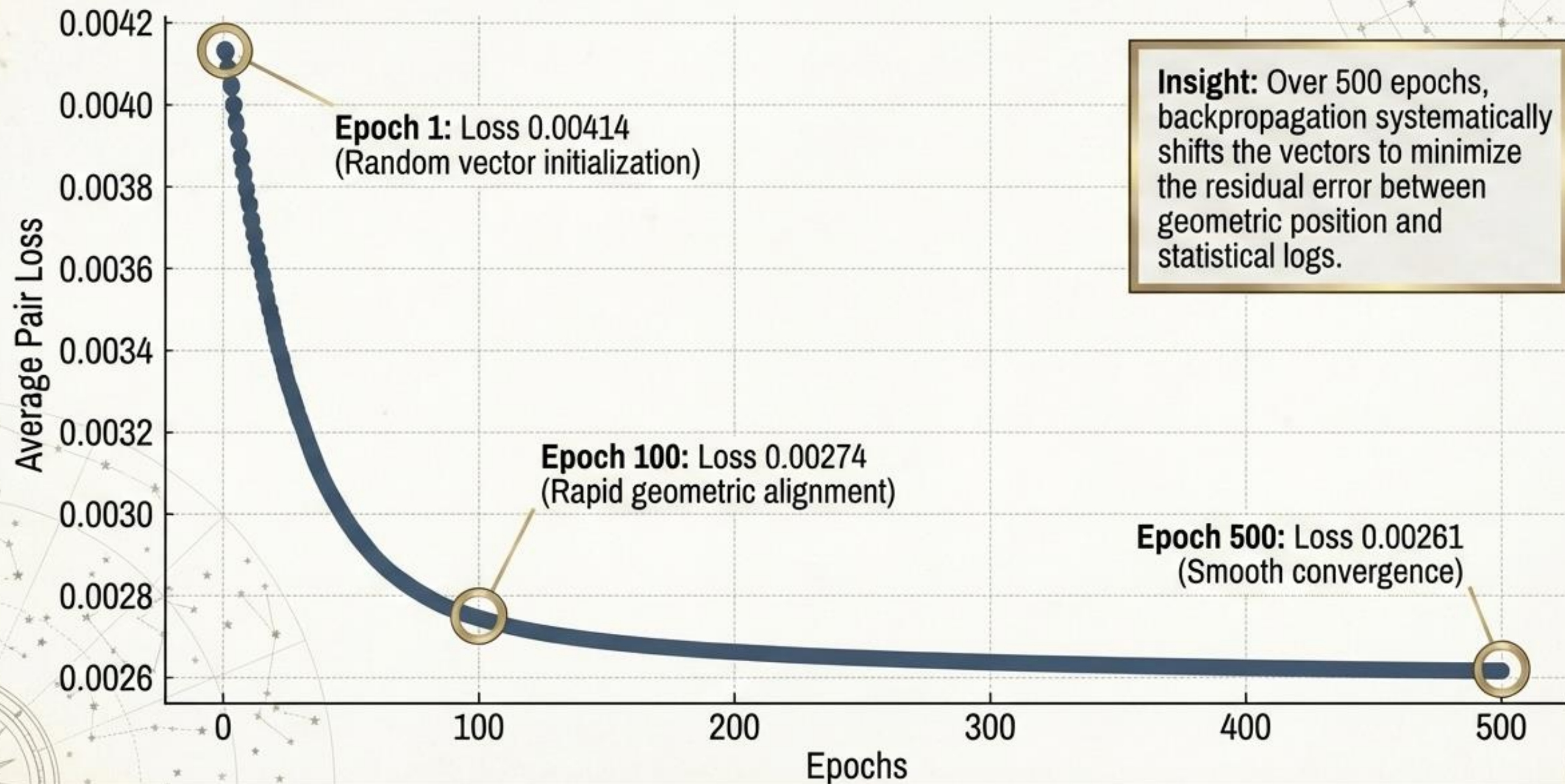
The Inverse-Distance Conveyor Belt



Resultant GloVe Neural Network (Single Training Pair)



GloVe Training Loss Over Epochs



Insight: Over 500 epochs, backpropagation systematically shifts the vectors to minimize the residual error between geometric position and statistical logs.



The 8-word sentence translated into **spatial coordinates**. Even in a microscopic corpus, **semantic pairs physically** drift toward each other to form continuous geometry.

Scaling Up: The Austen Probes

The mathematical principles are now applied to the 400,000-word GloVe Gigaword model.

We probe the pre-trained space using the thematic universe of Jane Austen to evaluate macro-scale semantic clustering.

NEAREST NEIGHBOR DATA

Probe	Neighbors
elizabeth	 - margaret (0.82) - mary (0.80) - jane (0.75)
darcy	 - bennet (0.54) - fitzwilliam (0.52)
marriage	 - divorce (0.84) - marry (0.71)
sense	 - feeling (0.78) - reason (0.75)

The Austen Galaxy

250 of the most frequent words projected into a 2D plane.

Observation: Connectors cluster centrally based on grammatical roles. Institutions and time form distinct outer bands.

The statistical structure of language is now visible geometry.

Focused t-SNE of Selected Jane Austen Words (GloVe)



Meaning is Inferred, Not Defined.

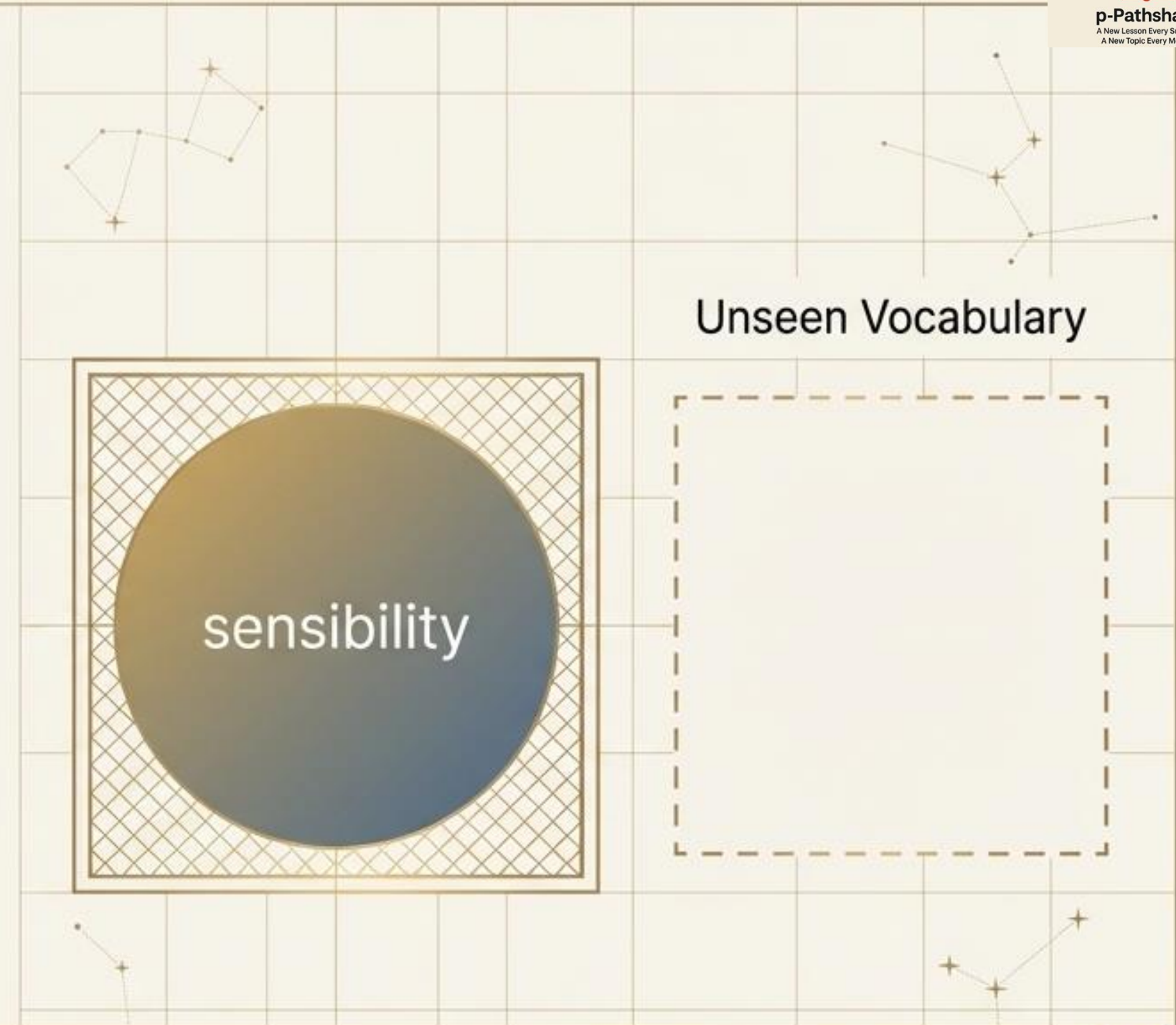
- The model possesses no dictionary, rules, or semantic constraints.
- It strictly enforces that vector differences mirror the logarithms of co-occurrence probabilities.
- Distance encodes similarity. Direction encodes relationships. Meaning is geometric.

The GloVe Limitation

GloVe captures meaning through global context, but treats every word as a rigid, indivisible atom.

The Unsolved Problems

- It ignores internal morphology (prefixes, suffixes, roots).
- It fails entirely on out-of-vocabulary words.



The Next Frontier: To handle true linguistic variation, language models must move inside the word. From global statistics to character n-grams. The geometric journey continues with FastText.

Support Independent Research: Fueling Discovery

The Financial Barriers to Research



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